



BIT1123/BISE2093/DIT1113

BIT1123 OBJECT ORIENTED PROGRAMMING

ASSIGNMENT 1 - INDIVIDUAL

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ALL SECTIONS

FACULTY OF INFORMATION TECHNOLOGY

CITY UNIVERSITY MALAYSIA

CYBERJAYA CAMPUS

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1. Introduction

Over the semester, I took a series of practical classes in Object Oriented Programming (OOP) on java. The tutorials helped me to enhance my programming skills by doing practical tasks as well as just learning theory. I was a little familiar with programming at the start of the semester, but I had some apprehensions in writing Java programs and in applying OOP concepts. Tutorials were progressively added to introduce key concepts in Java programming – variables, data types, conditional statements, loops, methods, arrays, ArrayLists, classes, objects, constructors, inheritance, encapsulation, polymorphism and other OOP concepts. I not only programmed, but I also learned to organize my work with the help of GitHub, to have a structured repository, and to document my work. Generally, I feel that the tutorials have helped me to grow in my technical programming skills as well as understanding of how software projects are undertaken and managed.

2. Knowledge Acquired by the Student from each Tutorial

My knowledge of Java and OOP has been enriched in various ways by each tutorial. The previous tutorials were about the basic components of Java programming including variables, data types, operators, output and input. These concepts were significant because they served as the basis for more advanced concepts that were to be discussed later. I then read up on decision making with if, else if, else. This helped me to see how programs can make decisions according to various circumstances. I also discovered the looping concepts of for and while, which enabled me to execute the same operations multiple times without repeatedly typing the same code. Another aspect was methods. I learned to break a program into smaller, more manageable pieces by writing methods to perform specific tasks. This greatly simplified the understanding, testing and maintenance of my programs. Also, I learned about the use of arrays and ArrayList to hold and manipulate a collection of data. The emphasis on the tutorials gradually changed to OOP. I learned how to define and use classes and objects and how to use attributes and methods in a class. Constructors helped me to understand that one could initialize them when they are getting created. I also learned about encapsulation, inheritance and polymorphism, which helped me to understand how larger java programs can be organized.

3. Challenges Encountered

The major problem I faced was to learn the syntax and structure of java programs. Programs may not compile if there are any small errors, like a missing semicolon, incorrect brackets, wrong type of variables, or a spelling error with a variable or method name. The other problem was to grasp the difference between classes and objects. I was struggling to understand why is it that a class is needed and how is an object created from a class. Constructors, inheritance, and polymorphism were also difficult to grasp as they made me think about programming in a more structured and Object-Oriented approach. Another problem I had with collections was with ArrayList. Accessing, adding and removing elements and looping through them needed some practice. Furthermore, fixing programs was sometimes tricky because I didn't always know right away what was wrong with my logic as indicated by the error message. Another challenge was maintaining my tutorial files and organization. The more exercises I added, the more important it became to make sure that my files had the correct names and were in the right folders for them to be uploaded to GitHub.

4. Answer the question: How did the challenges get overcome?

The main way I was able to overcome these difficulties was through repeated practice and debugging. If my program gave an error, I would read the compiler's error message and go and check the line that caused the error. Rather than flogging the old code, I tried to figure out why the error was occurring. Also looked at some examples from tutorials and tried out some minor changes to the programs. If things were tricky in OOP, I used to make small examples first, and then using these concepts in larger programs. For instance, I learned how to build a basic class containing attributes, constructors, and methods, before I got to work with multiple classes. Yet another technique that was beneficial to me was decomposition of the larger programming problems into smaller ones. Instead of doing the whole course all in one, I added a bit of a feature at a time, and tested it out before adding the next. This was helpful for debugging and for understanding how the different components of the program were related. GitHub also helped me to improve my organization. Having separate folders for each tutorial and making regular submissions allowed me to monitor my progress and have a good record of my progress over the semester.

5. Enhancement in Java skills.

During the tutorials, my java programming skills had been enhanced significantly. I started to get more used to Java syntax and to use variables, operators, conditions, loops and methods. I also learned to be a better programmer and to debug program errors. I think a big thing I learned was to break up a program rather than putting everything in one place to write all of the code. Methods could split various tasks, and classes could group related data and functionality. I made it better to work with collections like ArrayList, which is handy with multiple objects and fluctuating amount of data. I also became more confident in my ability to read and understand Java code, as I learned some of the common syntax and programming structures in Java. The biggest change I made was to become more comfortable in solving programming problems unaided. I can now analyse a problem, identify logic required and develop a solution step by step (with examples at least sometimes).

6. Knowledge of Object Oriented Programming Concepts

The tutorials greatly helped me to understand Object Oriented Programming. I got to know OOP (Object Oriented Programming) which is a style of programming that structures programs around objects that have data and behaviour. Encapsulation was about protecting the data of a class by restricting access to the data via methods like getters and setters. This aids to make packages safer, structured, and sustainable. I learned about inheritance's from it which was a concept of how one class can inherit properties and behaviours from another class. This helps to minimize the duplication of code and makes it possible for similar functionality to be shared between classes. Polymorphism was useful for me to grasp the idea of behavior changing based on the object calling the method or implementing the interface. This is helpful for developing flexible and reusable programs. I also got an understanding of classes and objects. Class is the blueprint and object are the instances that are created based on the blueprint. Constructors can then be used to put the data in when the object is created. In the overall, I changed my perspective of programming. I don't consider a program simply a list of instructions, I now see how various objects and classes can communicate with each other to form a whole.

7. Future Learning Plans

I would like to further develop my programming skills in java in the future by undertaking more challenging and meaningful projects. I want to be more confident in the use of advanced OOP concepts and designing programs with multiple interacting classes. I would also like to get more information about data structures, exception handling, file handling, databases and java frameworks in java. These will enable me to create more fully-fledged programs rather than just small tutorials. My other aim is to enhance my Git/GitHub skills. I want to know more about branching, pull request, conflict merging etc., because this is a part of the professional software development. I will also be creating a portfolio of a number of Java projects. Through programming practice-in conjunction with sufficient documentation and version control on GitHub-I can provide evidence of my technical growth and acquire skills that I will be able to use in future student projects and career.

8. Conclusion

Overall, it was an important part of my learning experience, the Java OOP tutorials. They not only taught me to become more proficient in Java programming, but they also gave me an understanding of some key software development concepts. I had some syntax errors, debugging, and troubles with collections, and some issues with OOP but through all this I did improve my problem-solving skills. The tutorials were very practical and gave me the opportunity to learn from mistakes and gain in confidence as I wrote the Java programs. Now I'm better versed in the notions of classes, objects, encapsulation, inheritance, polymorphism, methods and collections. I have also learned how to organize my work, and how to use GitHub as a version control and portfolio platform. In future, I plan to keep practicing java, create more complex projects, and enhance my OOP and professional software development skills. Overall, this learning experience has given me a good head start which will help me in future programming classes and projects.

9. GitHub Repository URL:

<https://github.com/basilmursi66-beep/BASEL-TAJELDEEN-MARSI-IDRIS-202504010013-OBJECT-ORINTED-PROGRAMMING>